

Modal Analysis of Uniform and Tapered Cantilever Beams

Project 01: Open FEM Project with FEniCSx

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1 Introduction and Physical Setup

Understanding the dynamic behavior of mechanical components is a standard requirement in structural engineering. When a structure is subjected to dynamic operational loads, it is necessary to ensure that the excitation frequencies do not coincide with the structure's natural frequencies to prevent resonance and potential fatigue failure.

This project investigates the modal characteristics (natural frequencies and mode shapes) of cantilever beams. The physical setup consists of a three-dimensional continuous solid domain anchored fixedly at one boundary (the root) and left completely unconstrained on all other boundaries. The objective is to identify the free vibration modes of the system under the assumption of small deformations and linear elastic isotropic material behavior.

To explore the influence of geometry on dynamic response, two distinct physical setups are considered:

1. **Variant A (Uniform Beam):** A standard cantilever beam with a constant rectangular cross-section along its longitudinal axis.
2. **Variant B (Tapered Beam):** A cantilever beam whose cross-section linearly decreases (tapers) from the root to the free end.

Figure 1 provides a visual comparison of the two variants.

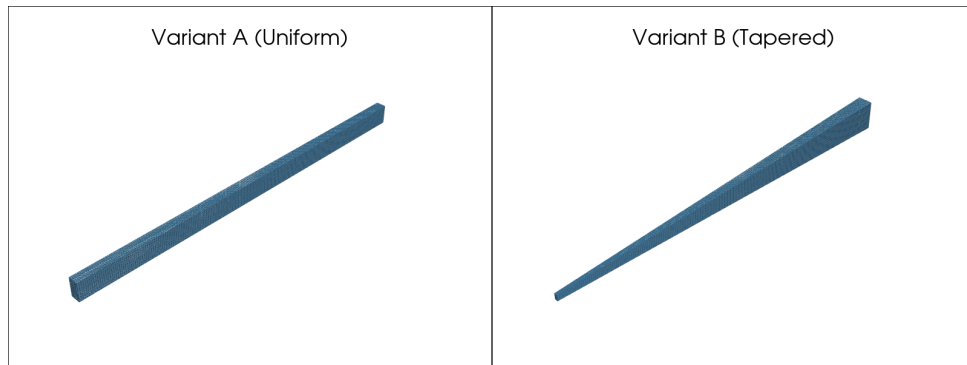


Figure 1: Geometric models used for the modal analysis: Variant A (Uniform Beam, left) and Variant B (Tapered Beam, right).

By analyzing these two variants, it is possible to systematically study how redistributing mass and stiffness affects the fundamental frequencies, an approach often employed in engineering to optimize natural frequencies while minimizing weight.

2 Mathematical Model and Weak Formulation

2.1 Strong Form

We consider a linear elastic body occupying an open, bounded domain $\Omega \subset \mathbb{R}^d$ ($d = 2$ or 3). The governing equation for the free vibration of an undamped structural system is given by the

elastodynamics equation without body forces [7]:

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} - \nabla \cdot \boldsymbol{\sigma}(\mathbf{u}) = \mathbf{0} \quad \text{in } \Omega, \quad (1)$$

where $\mathbf{u}(\mathbf{x}, t)$ is the displacement vector field, ρ is the mass density, and $\boldsymbol{\sigma}(\mathbf{u})$ is the Cauchy stress tensor.

Assuming a time-harmonic solution of the form $\mathbf{u}(\mathbf{x}, t) = \boldsymbol{\phi}(\mathbf{x})e^{i\omega t}$, where $\boldsymbol{\phi}(\mathbf{x})$ is the spatial mode shape and ω is the angular natural frequency, the temporal derivative yields $\frac{\partial^2 \mathbf{u}}{\partial t^2} = -\omega^2 \boldsymbol{\phi}e^{i\omega t}$. Substituting this into the governing equation and dropping the time-dependent term gives the stationary strong form of the eigenvalue problem:

$$-\nabla \cdot \boldsymbol{\sigma}(\boldsymbol{\phi}) = \lambda \rho \boldsymbol{\phi} \quad \text{in } \Omega, \quad (2)$$

where $\lambda = \omega^2$ is the eigenvalue.

For an isotropic linear elastic material, the stress tensor is related to the infinitesimal strain tensor $\boldsymbol{\varepsilon}(\boldsymbol{\phi}) = \frac{1}{2} (\nabla \boldsymbol{\phi} + (\nabla \boldsymbol{\phi})^T)$ [6] via Hooke's Law [3]:

$$\boldsymbol{\sigma}(\boldsymbol{\phi}) = 2\mu \boldsymbol{\varepsilon}(\boldsymbol{\phi}) + \lambda_L \text{Tr}(\boldsymbol{\varepsilon}(\boldsymbol{\phi})) \mathbf{I}, \quad (3)$$

where μ and λ_L are the Lamé parameters, related to Young's modulus E and Poisson's ratio ν by $\mu = \frac{E}{2(1+\nu)}$ and $\lambda_L = \frac{E\nu}{(1+\nu)(1-2\nu)}$.

2.2 Boundary Conditions

The boundary of the domain $\partial\Omega$ is decomposed into a Dirichlet boundary Γ_D and a Neumann boundary Γ_N , such that $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$.

- **Dirichlet Boundary (Γ_D):** The root of the cantilever is fixed, implying zero displacement:

$$\boldsymbol{\phi} = \mathbf{0} \quad \text{on } \Gamma_D. \quad (4)$$

- **Neumann Boundary (Γ_N):** All other surfaces are traction-free:

$$\boldsymbol{\sigma}(\boldsymbol{\phi}) \cdot \mathbf{n} = \mathbf{0} \quad \text{on } \Gamma_N, \quad (5)$$

where $\mathbf{n}$ is the outward-pointing unit normal vector.

2.3 Weak Formulation

To derive the weak form, the strong form is multiplied by a smooth vector-valued test function $\mathbf{v} \in V$ and integrated over the domain Ω :

$$-\int_{\Omega} (\nabla \cdot \boldsymbol{\sigma}(\boldsymbol{\phi})) \cdot \mathbf{v} \, dx = \lambda \int_{\Omega} \rho \boldsymbol{\phi} \cdot \mathbf{v} \, dx. \quad (6)$$

Applying integration by parts (Green's first identity) to the stress divergence term yields:

$$\int_{\Omega} \boldsymbol{\sigma}(\boldsymbol{\phi}) : \nabla \mathbf{v} \, dx - \int_{\partial\Omega} (\boldsymbol{\sigma}(\boldsymbol{\phi}) \cdot \mathbf{n}) \cdot \mathbf{v} \, ds = \lambda \int_{\Omega} \rho \boldsymbol{\phi} \cdot \mathbf{v} \, dx. \quad (7)$$

Exploiting the symmetry of the stress tensor, $\boldsymbol{\sigma} : \nabla \mathbf{v} = \boldsymbol{\sigma} : \boldsymbol{\varepsilon}(\mathbf{v})$. Splitting the boundary integral into Γ_D and Γ_N :

$$\int_{\Omega} \boldsymbol{\sigma}(\boldsymbol{\phi}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, dx - \int_{\Gamma_D} (\boldsymbol{\sigma}(\boldsymbol{\phi}) \cdot \mathbf{n}) \cdot \mathbf{v} \, ds - \int_{\Gamma_N} (\boldsymbol{\sigma}(\boldsymbol{\phi}) \cdot \mathbf{n}) \cdot \mathbf{v} \, ds = \lambda \int_{\Omega} \rho \boldsymbol{\phi} \cdot \mathbf{v} \, dx. \quad (8)$$

The test function $\mathbf{v}$ is chosen to vanish on the Dirichlet boundary Γ_D ($\mathbf{v} = \mathbf{0}$). Furthermore, the natural traction-free boundary condition enforces $\boldsymbol{\sigma} \cdot \mathbf{n} = \mathbf{0}$ on Γ_N . Thus, both boundary integrals vanish.

The weak form is: Find $(\lambda, \boldsymbol{\phi}) \in \mathbb{R} \times V$ such that

$$a(\boldsymbol{\phi}, \mathbf{v}) = \lambda m(\boldsymbol{\phi}, \mathbf{v}) \quad \forall \mathbf{v} \in V, \quad (9)$$

where the stiffness bilinear form $a(\cdot, \cdot)$ and the mass bilinear form $m(\cdot, \cdot)$ are defined as:

$$a(\boldsymbol{\phi}, \mathbf{v}) = \int_{\Omega} \boldsymbol{\sigma}(\boldsymbol{\phi}) : \boldsymbol{\varepsilon}(\mathbf{v}) \, dx, \quad (10)$$

$$m(\boldsymbol{\phi}, \mathbf{v}) = \int_{\Omega} \rho \boldsymbol{\phi} \cdot \mathbf{v} \, dx. \quad (11)$$

The suitable function space for both trial and test functions is:

$$V = \{\mathbf{v} \in [H^1(\Omega)]^d \mid \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D\}. \quad (12)$$

3 Implementation and Solver Choices

The project is implemented using the **FEniCSx** ecosystem [8]. The work was elaborated as a Python project hosted and deployed via Git following the instructional structural guidelines [2]. The complete source code and implementation details can be found in the project repository [1]. Next, the software structure is detailed.

3.1 Software Stack

- **Meshing:** Geometries are programmatically defined and meshed using the **Gmsh** Python API (`gmsh.model OCC`) [11]. The meshes are imported into DOLFINx using `dolfinx.io.gmshio.read_from_msh`.
- **Finite Elements:** Continuous Lagrange finite elements of degree 1 or 2 (P_1 or P_2) on tetrahedral or hexahedral meshes, provided by **basix** and **ufl** [8].
- **Assembly:** The bilinear forms $a(\boldsymbol{\phi}, \mathbf{v})$ and $m(\boldsymbol{\phi}, \mathbf{v})$ are assembled into PETSc matrices

using `dolfinx.fem.petsc` [10]. Essential boundary conditions are applied to the assembled stiffness matrix algebraically.

- **Eigensolver:** The generalized eigenvalue problem $K\phi = \lambda M\phi$ is solved using **SLEPc** (Scalable Library for Eigenvalue Problem Computations) via `slepc4py` [9]. The Krylov-Schur algorithm with a shift-and-invert spectral transformation is utilized to compute the smallest magnitude eigenvalues [5].
- **Post-processing:** ParaView outputs (in XDMF/VTK formats) were generated to visualize the computed mode shapes and deformation fields.

4 Verification and Mesh Convergence Study

4.1 Verification by Analytical Benchmark

To verify the solver, the eigenfrequencies of a 3D uniform cantilever beam are computed and compared against the 1D analytical solutions provided by the Euler-Bernoulli beam theory [4]. For a uniform beam of length L , cross-sectional area A , area moment of inertia I , Young's modulus E , and density ρ , the analytical natural frequencies (in rad/s) are:

$$\omega_{i,\text{analytical}} = \beta_i^2 \sqrt{\frac{EI}{\rho AL^4}}, \quad (13)$$

where β_i are the roots of the characteristic equation $\cos(\beta) \cosh(\beta) = -1$.

4.2 Mesh Convergence Study

A mesh convergence study was performed to check spatial discretization accuracy. The study utilized continuous Lagrange elements and progressively refined the mesh resolution $h = L/N_x$, where N_x is the number of elements along the longitudinal axis.

The relative error for the first eigenfrequency is defined as:

$$\text{Error}(h) = \left| \frac{f_{1, \text{FEM}}(h) - f_{1, \text{analytical}}}{f_{1, \text{analytical}}} \right|. \quad (14)$$

Figure 2 shows the results of the mesh convergence study plotted on a log-log scale.

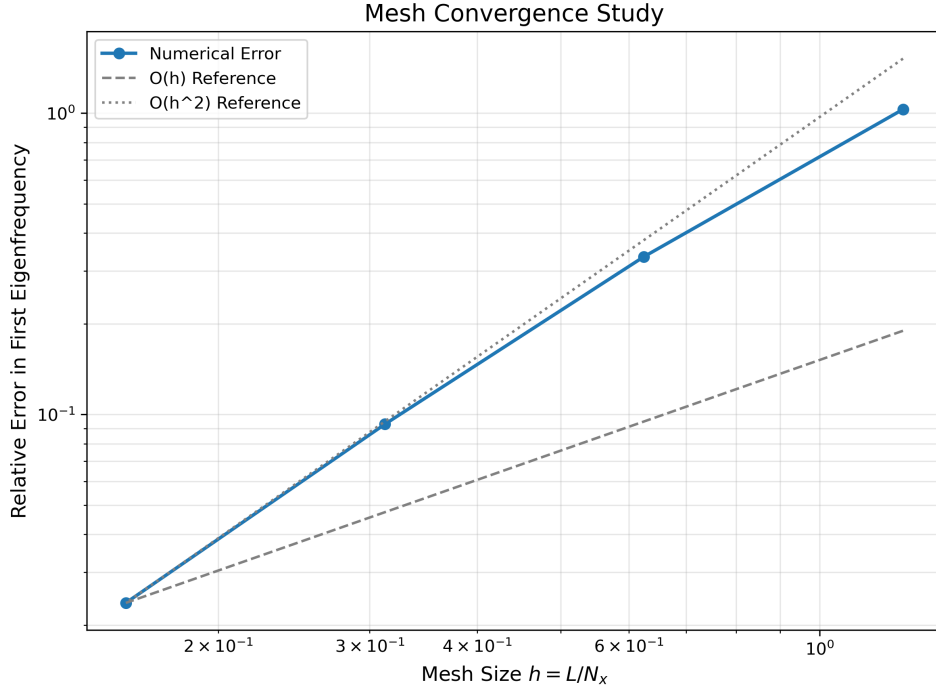


Figure 2: Mesh convergence study for the first eigenfrequency of the uniform cantilever beam. The plot shows the relative error versus mesh size h on a log-log scale.

As observed in Figure 2, the numerical error decreases as the mesh size h is reduced. The experimental convergence rate aligns with the $\mathcal{O}(h^2)$ reference line, which is expected for eigenvalue problems solved with linear (P_1) finite elements.

5 Realistic Scenario Study: Uniform vs. Tapered Cantilever

We deploy the FEM solver to study a structural scenario: modifying the mass distribution by tapering the beam.

5.1 Geometry and Setup

We compare two variants:

- **Variant A (Uniform):** A standard rectangular beam, $L = 1.0\text{m}$, constant cross-section $0.1\text{m} \times 0.1\text{m}$.
- **Variant B (Tapered):** A beam of the same length $L = 1.0\text{m}$, with a root cross-section of $0.1\text{m} \times 0.1\text{m}$, but linearly tapering in both transverse directions to a tip cross-section of $0.02\text{m} \times 0.02\text{m}$.

5.2 Analysis of Quantities of Interest

The primary quantities of interest are the fundamental eigenfrequencies and their corresponding mode shapes (Modes 1 through 6).

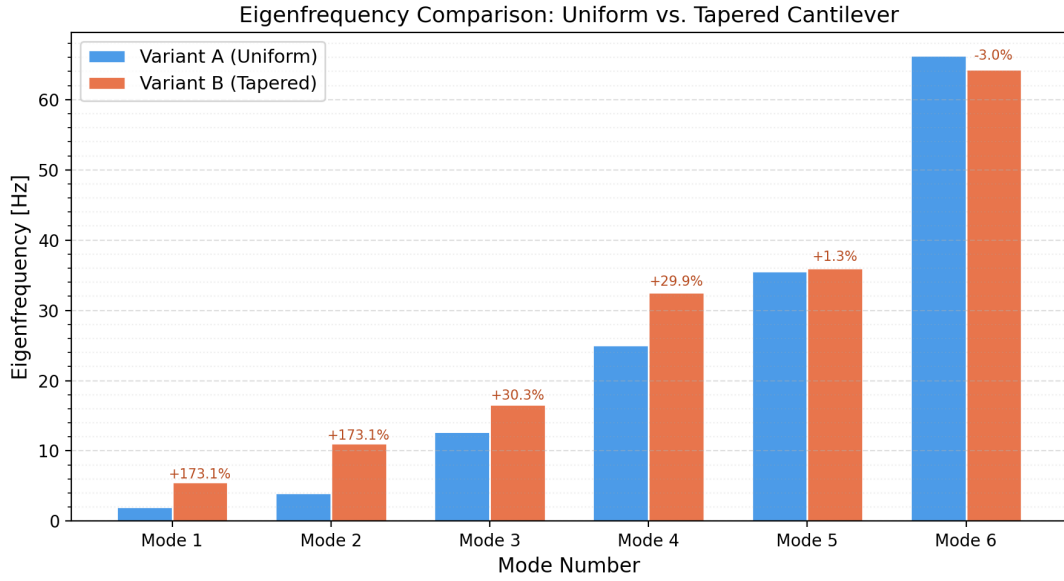


Figure 3: Comparison of the first six eigenfrequencies for Variant A (Uniform) and Variant B (Tapered). Percentage labels indicate the relative shift from Variant A to Variant B.

Figure 3 illustrates the computed eigenfrequencies for both variants. The results show differences in the dynamic response:

1. **Fundamental Bending Modes (Modes 1 and 2):** The tapered beam shows a **+173.1% increase** in its first two fundamental frequencies. Tapering removes mass primarily from the tip while preserving most of the bending stiffness at the root.
2. **Higher-Order Bending Modes (Modes 3 and 4):** The second bending mode pair also sees an increase (+30.3% and +29.9%).
3. **Torsional/Longitudinal Modes (Modes 5 and 6):** Mode 5 sees a minimal change (+1.3%), and Mode 6 experiences a slight **decrease (-3.0%)**. Torsional stiffness decreases with a reduced cross-section, which contract the loss of mass.

5.3 Mode Shape Visualization

To further understand the modal behavior, the displacement fields for the first six modes were visualized.

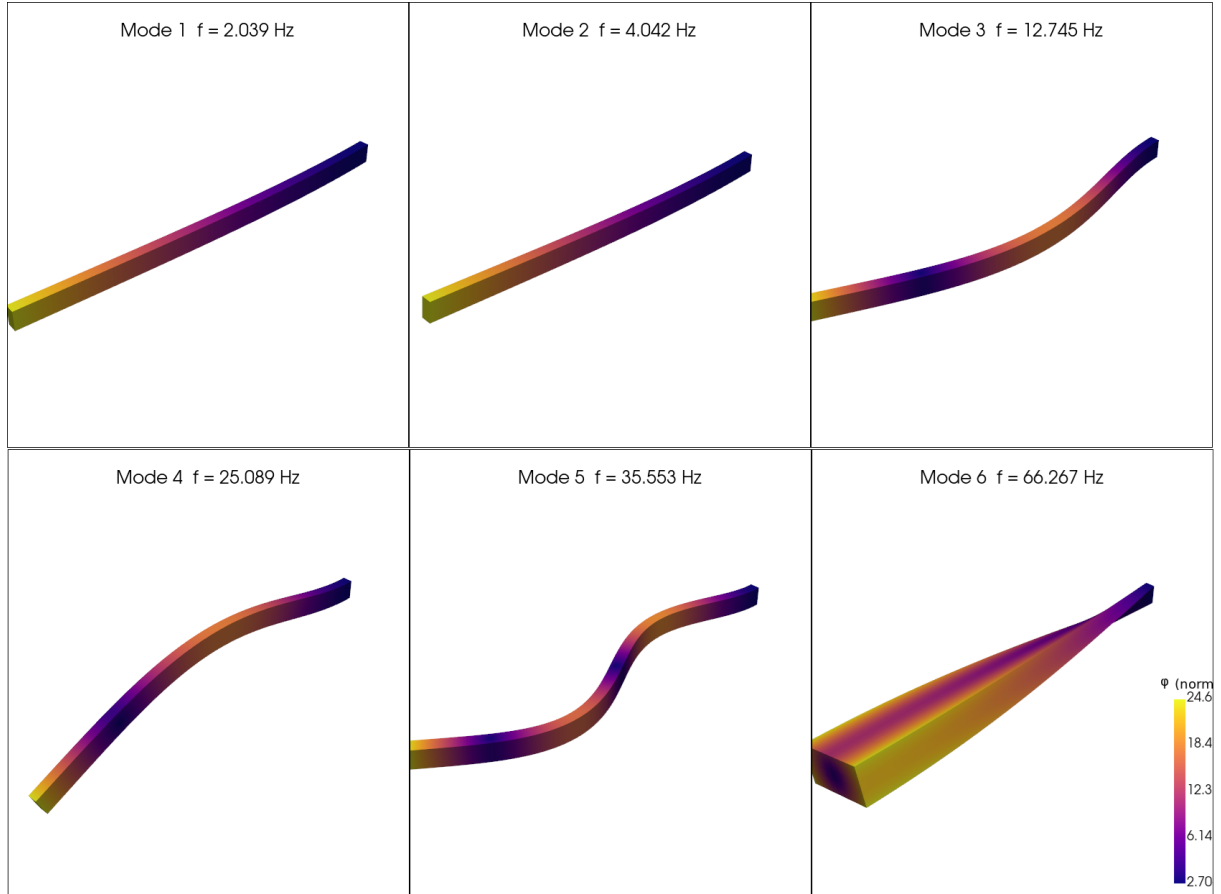


Figure 4: First six mode shapes of the Uniform Cantilever Beam (Variant A). Displacements are exaggerated for clarity. Colors indicate displacement magnitude.

Figure 4 displays the mode shapes for Variant A. As expected for a symmetric uniform beam, Modes 1 and 2 represent the fundamental bending modes along the two principal axes of the cross-section. Modes 3 and 4 are the second-order bending modes, characterized by an intermediate nodal point (a point of zero displacement along the beam length). Modes 5 and 6 exhibit more complex behavior, typically involving torsion or longitudinal elongation.

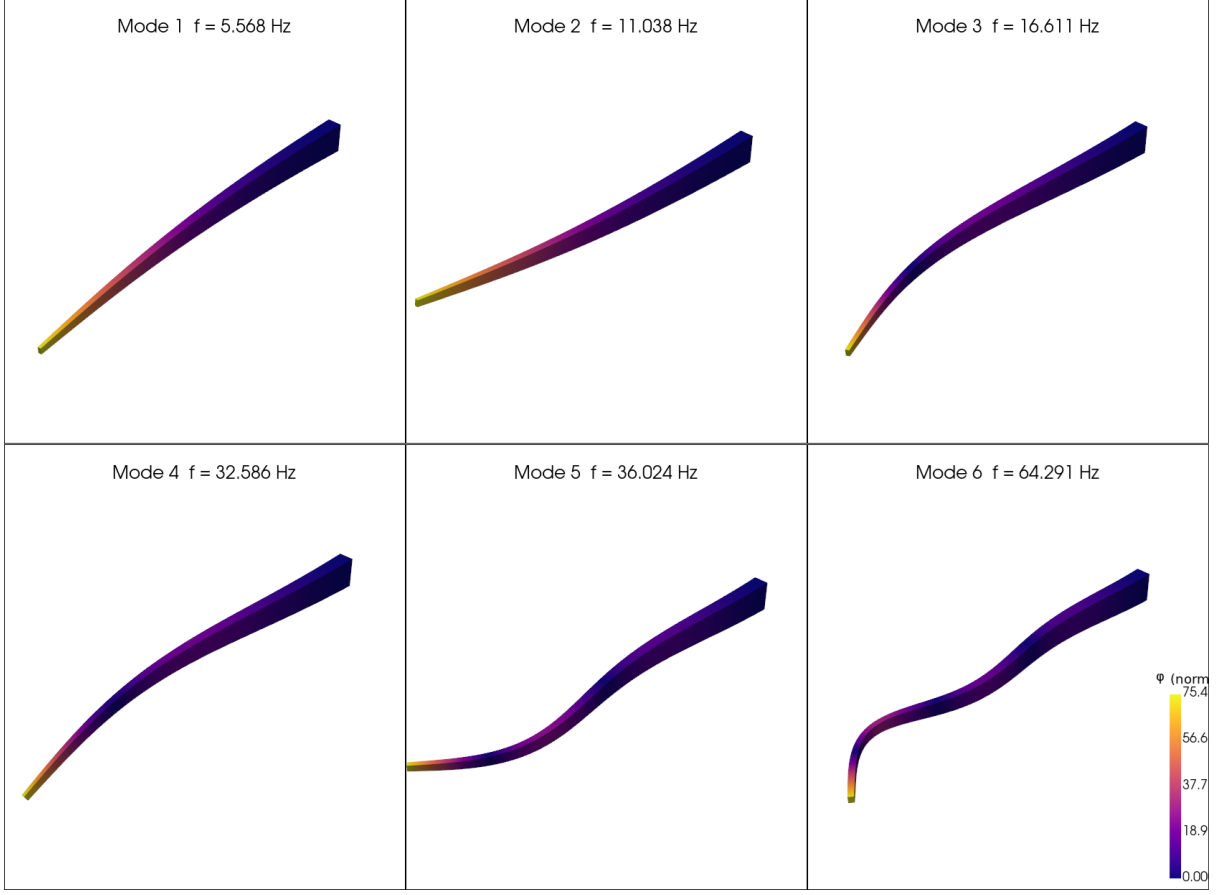


Figure 5: First six mode shapes of the Tapered Cantilever Beam (Variant B). Displacements are exaggerated for clarity. Colors indicate displacement magnitude.

Figure 5 displays the mode shapes for Variant B. While the general sequence of mode types (fundamental bending, second-order bending, etc.) remains similar to Variant A, the spatial distribution of the deformation is altered. The tapering concentrates the higher deformation gradients towards the thinner tip of the beam, which has lower local stiffness compared to the uniform case. This visualizes why the mass reduction at the tip has such a pronounced effect on the lower-order bending frequencies.

6 Conclusion

A finite element solver for structural modal analysis was implemented using FEniCSx and SLEPc. The derivation of the elastodynamics weak form accounted for Dirichlet and Neumann boundary conditions.

Verification against the analytical Euler-Bernoulli beam theory showed good agreement, and the mesh convergence study confirmed the theoretical $\mathcal{O}(h^2)$ convergence rate for the eigenfrequencies.

The application of the solver to compare a uniform beam to a tapered beam provided physical insights. The analysis indicated that linear tapering can shift fundamental resonances higher

(+173.1%) while reducing material volume, though it can slightly lower some higher-order torsional mode frequencies.

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